

DHARANI SUNDHARAM G

Hardware Engineer

Electronics and Communication Engineering student with a multifaceted background in **Digital Logic**, **Autonomous Systems**, and AI-driven architectures. Experienced in developing secure agentic operating systems and assistive hardware at **dVerse Technologies**. A versatile problem-solver with an elite background in security research, used to build robust and reliable hardware-software interfaces.



WORK EXPERIENCE

dVerse Technologies (iTNT Hub)

Mar 2025 - Present

Hardware and Software Specialist (Freelancer).
Product Development Intern

- Engineered **FlowAX**, an **agentic** operating system utilizing local LLMs and vector databases like **RAG** for secure, high-performance document processing [User Summary].
- Development of the intelligent counseling system for Higher Secondary Students
- Developed **hardware-software** integration for the **CuriO** tactile tablet, optimizing real-time feedback and accessibility features for visually impaired users.
- Implemented secure **IoT communication protocols** for STEM education applications.

PROJECTS

Digital Systems & Robotics Hardware Developer

- Designed a **4-bit arithmetic processor** using discrete logic components, demonstrating core knowledge of RTL design and digital logic.
- Engineered a **Vehicle-to-Vehicle (V2V)** platform using Raspberry-pi and **ROS** based **SLAM** and integrated collision avoidance.
- Developed high-precision motor control systems for **autonomous** robots using AccelStepper and **PID** logic.

Systems Research & Security Penetration Tester

- Conducts research in IoT and **Reverse Engineering**, focusing on automated vulnerability assessments.
- Maintains a **Top 100 Global Ranking** on "Hack The Box" and consistently ranks in the **Top 10 in India** for national CTF competitions.
- Developed **Blackeye**, an automation tool for streamlining ethical hacking and system administration workflows

CONTACT

Phone

+91 9626262428

Email

dharani3318s@gmail.com

Address

Ambattur, Chennai-53

Website

dharanisundharam.vercel.app

EDUCATION

2024 - 2028

R.M.D ENGINEERING COLLEGE

- Bachelor's in Engineering
Electronics and
Communication
Engineering

SKILLS

- Digital Design:** Verilog HDL, RTL Design, **FPGA Basics**.
- Systems & AI:** Local LLMs, RAG Pipelines, **Linux OS**, **Docker**.
- Programming:** Python, Java, C++, Bash Scripting.
- Hardware/IoT:** **PCB Design (KiCad)**, ESP32, Raspberry Pi, **STM32**.
- Robotics:** V2V Systems, Motor Control, **Autonomous Navigation**, **Industry Robots (CRA5)**.
- Security:** Defensive security research and **vulnerability assessment** (HTB Global Top 100).